MCS Saturday SAT Math Homework

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Linear Expressions + Equations

Digital SAT Math · Session 3

Distribute Fully · Isolate Precisely · Homework Activity Sheet

Name _____ **Date** _____

Completion Check Score after review; justify every correction in the error log.	____ / 18 Core	____ / 6 Bridge	____ / 24 Total

A Structure Before Symbols

4 prompts · 7 min

Do not solve unless asked. Name the legal move and explain what it preserves.

A1 · Distribution — Explain

For $-4(2x + 7)$, describe exactly how the factor -4 changes **both** terms. Why must the constant change sign?

State the two products and the sign logic.

A2 · Like Terms — Classify

From $5x^2, -3x, 8, 11x^2, 2x$, list the groups that may legally combine. Explain why x^2-terms and x-terms are not alike.

Group first; then defend the boundary.

A3 · Equality — Audit

A student changes $6x - 11 = 31$ to $6x = 31$. Identify the first invalid step and write the correct next line.

Name what must happen to both sides.

A4 · Invariant — Explain

In one precise sentence, distinguish **rewriting one side equivalently** from **performing an operation on both sides**.

Use the phrase “solution set.”

B Translate and Defend

4 prompts · 9 min

Define the variable with units, write an equation, solve, and interpret the result.

B1 · Context Model — Construct

A bicycle rental costs a one-time safety fee of **$9** plus the same hourly rate. A 6-hour rental costs **$57**. Let **r** be the hourly rate in dollars per hour. Write and solve an equation for **r**.

Define → model → solve → check units.

Hourly rate: _____

B2 · Meaning — Explain

In your equation from B1, explain which amount occurs once, which amount repeats, and why the coefficient is 6.

B3 · Representation — Choose

Which equation represents: “Seven less than three times a number is 20”?

A. $7 - 3x = 20$

B. $3(x - 7) = 20$

C. $3x - 7 = 20$

D. $7x - 3 = 20$

B4 · Distractor Autopsy — Analyze

For B3, choose one incorrect option and describe the exact language error that would produce it.

Do not merely say “wrong order.” Connect words to symbols.

C Linear Fluency Lab

8 questions · 14 min

Write one legal move per line. Substitute every answer into the original equation before moving on.

Distribute completely · Combine within each side · Create one variable side · Plug back

C1 Level 1

8x + 13 = 61

x = _____

C2 Level 1

4(3x − 2) = 52

x = _____

C3 Level 2

15 − 2(4x + 1) = −19

x = _____

C4 Level 2

7x + 18 = 3x − 10

x = _____

C5 Level 2

5(2x + 3) − 4 = 3(x + 9)

x = _____

C6 Level 3

4 − 3(2x − 5) = 5(x + 1) + 3

x = _____

C Linear Fluency Lab (continued)

8 questions · 14 min

Write one legal move per line. Substitute every answer into the original equation before moving on.

Distribute completely · Combine within each side · Create one variable side · Plug back

C7 Classify

Determine whether the equation has **one solution, no solution, or infinitely many solutions**.

$6(x - 2) + 4 = 2(3x - 4)$

Classification: _____

C8 Classify

Determine whether the equation has **one solution, no solution, or infinitely many solutions**.

$9 - 4(2x + 1) = 5 - 8x$

Classification: _____

D Equivalent or Not?

3 audits · 6 min

For each proposed step, mark Valid or Invalid. If invalid, repair only that step.

D1 · Distribution Audit

$-3(5x - 4) \rightarrow -15x - 12$

A. Valid **B.** Invalid

Repair or justify.

D2 · Both Sides Audit

$2x + 9 = 17 \rightarrow 2x = 8$

A. Valid **B.** Invalid

Name the operation.

D3 · Combining Audit

$4x^2 + 7x - 3x^2 \rightarrow 8x^2$

A. Valid **B.** Invalid

Rewrite the expression correctly and explain the variable parts.

E Guardrail Bridge

6 questions · 12 min

Use the shortcut only when its condition is satisfied. For radicals, generate candidates and then return to the original equation.

E1 · Proportion — Solve

(x + 5)/6 = (3x − 1)/10

x = _____

E2 · Permission — Explain

A student wants to “cross-multiply” in **(x + 2)/3 + 4 = 7**. Explain why the diagonal shortcut does not apply as written, then give a legal first move.

E3 · Radical — Solve + Check

√(3x + 4) = x

Valid solution(s): _____

E4 · Domain — Reason

Before squaring **√(5x − 6) = x − 4**, what two conditions must be true? State each as an inequality.

Check the radicand and the right side.

E5 · Candidate Trial — Verify

Algebra produces candidates **x = 2** and **x = 7** for **√(x + 2) = x − 4**. Test both in the **original** equation and report the verdict.

E6 · Error Mechanism — Explain

Why can squaring both sides create a solution that was not valid before squaring? Use sign information in your explanation.

F Digital SAT Timed Set

6 questions · 9 min

Set a 9-minute timer. Answer every item. For graph verification, identify the requested coordinate before opening Desmos.

Box the deliverable · Choose algebra or graph · Audit signs · Zero blanks

F1 Level 1

Which expression is equivalent to **3(4x − 5) + 2x**?

A. 14x − 5

B. 14x − 15

C. 12x − 13

D. 6x − 15

F2 Level 2

If **11 − 3(2x − 1) = −22**, what is the value of **x**?

SPR: _____

F3 Level 2

A meal-delivery plan charges a one-time cooler deposit of **$24** and **$11** per meal. A customer's first order costs **$101**. How many meals were ordered?

SPR: _____

F4 Level 3

For what value of **k** does **k(2x − 3) = 10x − 15** have infinitely many solutions?

A. 2

B. 3

C. 5

D. 15

F5 Level 3

The lines **y = 4x − 9** and **y = −2x + 15** intersect at **(a, b)**. What is the value of **a**?

Algebra or Desmos verification.

SPR: _____

F6 Level 4

If **(m + 2)x + 7 = 9x + 7** is true for every value of **x**, what is **m**?

SPR: _____

G Verification Record

Evidence before answers

Complete this section before checking an answer key or discussing solutions.

- ☐ I distributed to every term, including constants.
- ☐ I kept equations equivalent one line at a time.
- ☐ I stated conditions before using an advanced shortcut.
- ☐ I answered the requested quantity, coordinate, or unit.
- ☐ I combined only terms with identical variable parts.
- ☐ I checked each linear solution in the original equation.
- ☐ I tested every radical candidate in the original equation.
- ☐ I finished the timed set with zero blanks.

Timing Evidence Record

How long did F1–F6 take? Which question consumed the most time, and what observable trigger should have made you flag it?

Tool Evidence Record

If you used Desmos, state the two expressions entered, the graph feature inspected, and the exact value read. If not, identify one item it could verify.

H Error Log

Up to 5 corrections

Choose one primary error type: Distribution, Like Terms, Equality, Translation, Condition, Checking, Reading, or Timing.

#	Problem	My original thought	Primary error type	Correction rule
1				
2				
3				
4				
5				

I Transfer Reflection

Complete sentences

1. The first visible sign that I have made a distribution or sign error is…

2. One equation in this homework that I can now solve by two representations is… The representations are…

3. The distinction between a candidate and a verified solution matters because…

4. On my next Digital SAT Math module, the correction rule I will apply first is…